{EPITECH}

SMART FRIDGE_

< INITIATING AND PLANNING A PROJECT />



SMART FRIDGE

Modern houses are becoming increasingly connected as digital technologies integrate into everyday life.



Smart home devices now automate comfort, energy management, hygiene, security, and much more, while collecting data to improve user experience. They include thermostats, lighting, voice assistants, door locks and bells, vacuum cleaners, etc.









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A mid-sized home-appliance company is under pressure: traditional refrigerators still sell, but margins are shrinking, competition is intense, and innovation has slowed down.

The executive board believes the **smart connected fridge** could be the next growth lever.





However, before investing millions, they need answers. Is there a real user need? Does it make business sense? Is it technically and operationally feasible?

Because you're an expert in digital transformation, they tasked you with **framing, analyzing, and structuring the project** in order to support an informed go/no-go decision.

Your Mission

Your mission is to **evaluate and structure the launch of a smart connected refrigerator**, from early exploration to commercialization readiness.

You will not perform the actions to build the project, but rather define these actions and investigate the project's feasibility and viability.

You are in charge of:

- ✓ Framing the project;
- ✓ Presenting a production and delivery strategy;
- ✓ Investigating resources needed;
- ✓ Presenting a budget proposal;
- ✓ Ensuring risk containment;
- ✓ Identifying relevant skills and profiles;
- ✓ Communicating about the project.



Framing the Project: Scope and Expectations

Before moving into execution, the company wants to ensure the project is properly framed. At this stage, the objective is **not to design or validate the product itself**, but to define the conditions under which a relevant, feasible, and well-structured product could later be developed.

Beyond the usual market analysis, the organization lacks a clear approach for initiating a project. Your role is therefore to **structure the early ideation and conception phase** by defining the necessary steps.

You must clearly describe:

- ✓ which steps are required;
- ✓ why each step is important and what each step is meant to achieve;
- ✓ what common tools or methodology can be used to perform each step;
- ✓ the order in which the steps should occur and how they logically connect to one another.

The goal is to reduce uncertainty, align stakeholders, and prepare informed decision-making before committing significant time, budget, and resources.



You are expected to explain *why, how and what should be done before any development starts*, not to actually perform those activities.

Expected Deliverables

At the end of this framing phase, you are expected to deliver **a coherent and structured set of documents that define the approach, not the outcomes**:

- ✓ A **project framing document** that synthesizes the problem space, the initial assumptions, the scope of the project, and what is explicitly out of scope. This acts as a shared reference for all stakeholders.
- ✓ A **research plan** describing which analyses should be conducted later (market study, user research, competitive analysis), including their objectives, methods, and expected outputs, without executing them yet.
- ✓ A **high-level roadmap** describing the key steps of the ideation and conception process, their sequence, and their dependencies. This roadmap focuses on *what needs to be clarified*, and highlights what must be validated first and why.

These deliverables should demonstrate your ability to think upstream, structure ambiguity, and lay solid foundations for a complex product and project initiative.

Project Management Plan

Your next challenge is to structure the path to the product delivery, using a **Project Management Plan** that describes how the project would be organized, governed, and controlled if it were to move forward.
A single document would make your *PMP* hard to maintain and use. You must produce a lightweight master document that explains the overall project management approach and references a set of specialized supporting documents. Together, these documents form a coherent and consistent whole.

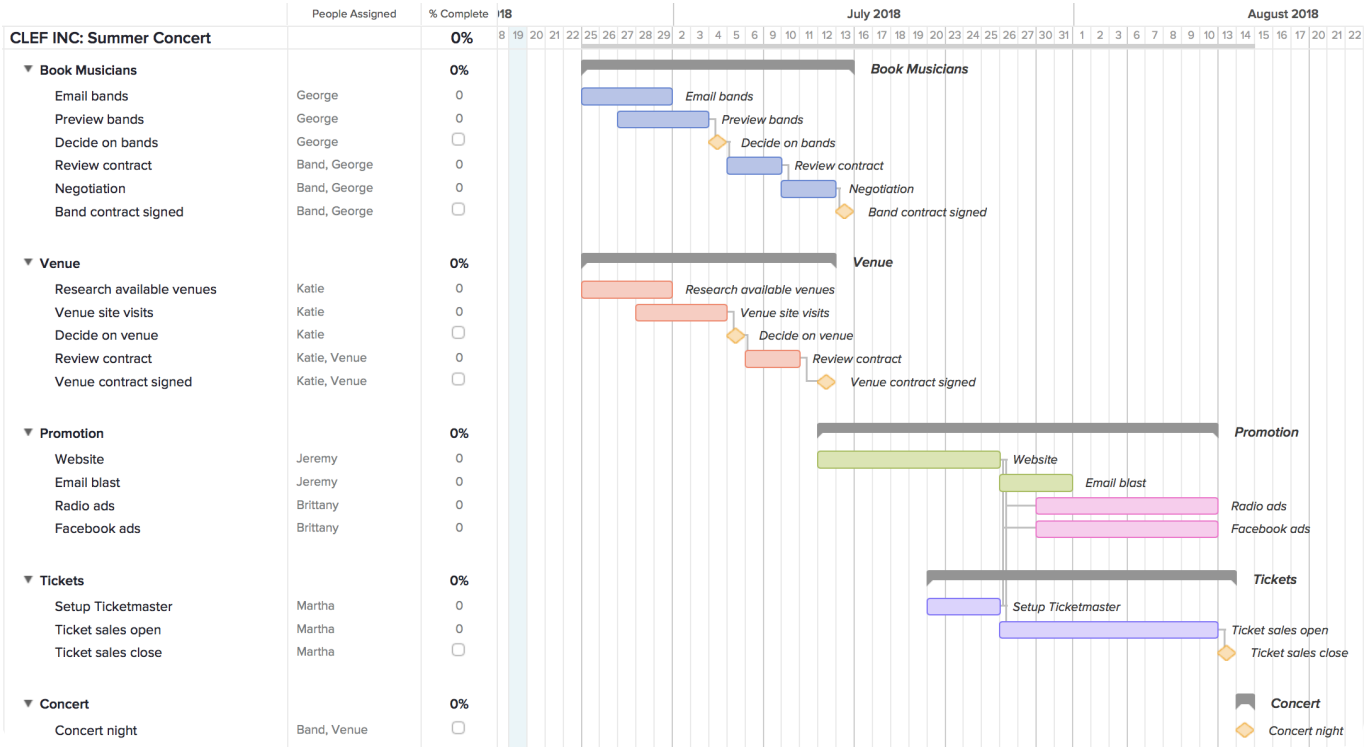


Take inspiration from proven industry project management frameworks (e.g., PMBOK, PRINCE2).

All these documents are iterative by nature and must be updated whenever new insights, constraints, or gaps are identified. Any modification should be easy to identify from your repository commits.

Time management

You have to plan how the project would unfold, from framing to delivery through implementation. You must submit a Gantt chart to structure phases and tasks, show dependencies, and make assumptions about sequencing, duration, and staffing. It must reflect decision points, parallel activities, and logical links. A first version should show your initial reasoning and assumptions, and it should be revised as the project progresses and new insights emerge.



Human

Effective delivery depends on having the right skills, roles, and mindsets at the right time. You are asked to think beyond tasks and consider the human structure required to carry the project forward, and define the profiles that would logically be involved in such a project.

Based on your project framing and planning, analyze the work that needs to be done in order to identify the **skills and profiles required**.

You are expected to submit a **profiles and skills document**, structured as a set of personas, with one persona per page. Each persona should describe a role, its main responsibilities, key skills, and its contribution to the project.

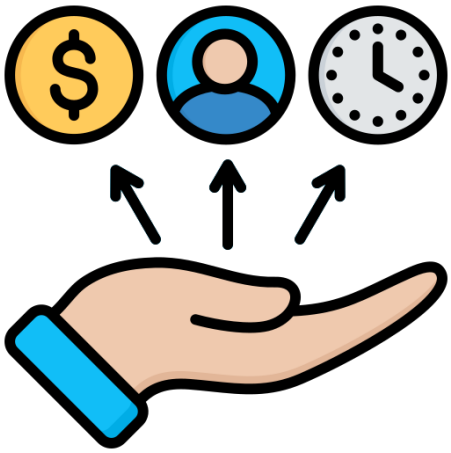
This should also help you test the coherence of your planning. If identifying skills reveals missing tasks or underestimated activities, you are expected to revisit and update your project plan accordingly. Any adjustment resulting from this analysis should be reflected in your Gantt chart.

Material

Beyond people, a project also relies on a set of material, technical, and organizational resources. You are asked to anticipate everything that would be required to carry the project forward: tools, technologies, equipment, services, trainings, workshops, or any other supporting elements.

Your objective is to think realistically about availability and constraints. Not all resources are instantly accessible, and some may introduce delays or dependencies that must be reflected in your planning. Any resource-related constraints or dependencies must be reflected in your Gantt chart.

You are expected to submit a **resource list**, structured as a table, identifying the required material resources and, where relevant, quantities and availability assumptions.



Cost management

Once skills and resources have been identified, you should be able to assess their financial impact in order to build a **budget projection** that reflects the cost of your assumptions.

You are expected to submit a **budget proposal**. Research realistic price ranges, organize costs into categories, and distinguish between one-off and recurring expenses. The goal is to demonstrate economic realism and an ability to justify choices.



This is not an accounting exercise, there is not a single correct solution. Your coherence and relevance will be evaluated, not the exactness.

Any budget deadlines and constraints must be reflected in your Gantt.

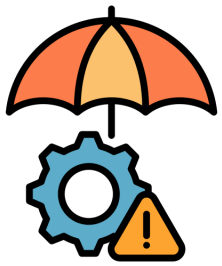
Risk management

No project is free from uncertainty. You're asked to step back and identify the main risks that could threaten the success of the project.

You are expected to submit a **risk register** identifying at least five major risks, with an assessment of their probability and impact, and a proposed mitigation strategy for each.

The focus is not on exhaustiveness, but on relevance: identifying the most critical risks and thinking through how they could be mitigated.

If a risk affects planning, budget, resources, or skills, the corresponding documents must be updated accordingly.



Communication management

Communication is a core part of project management. A project can only succeed if it is communicated clearly and appropriately.

You're asked to focus on two key audiences: the project team, and a supervisory or decision-making stakeholder. You must explain the project, its constraints, and its needs in a clear and professional way.

You are expected to produce:

- ✓ One **internal communication** addressed to the project team, presenting the project structure, objectives, and management approach;
- ✓ One **executive communication** addressed to a company board member, focused on the budget request and its justification.



📢 Adapt your communication to the audience, each with their own expectations, level of involvement, and technical expertise.

⚠️ The preparation of these communications must appear in your planning.

Presentation and Delivery

The delivery consists of a **review** of your documents, as a group.

You are expected to:

- ✓ Present how you validated the **technical feasibility**;
- ✓ Justify your choices, especially for resources and budget;
- ✓ Explain your methods for mitigating **market-related risks** (adoption, pricing, competition);
- ✓ Show how you adapted your **planning** and deliverables in response to unforeseen events.

You have to explain the pitfalls you encountered and how you addressed them.

⚠️ Your delivery must be properly structured to allow anyone to easily grasp the project.



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